LAB WORKSHEET

Network Design Lab

Turn a client brief into a justified network design with a topology and a VLSM addressing plan.

Process

1. Read the brief: users, devices, growth, security and availability needs.
2. Draw the topology: edge firewall, switches, APs, servers, end devices; choose link types; draw zones.
3. Assign VLANs per department, plus guest, voice and management.
4. Plan addresses with VLSM, largest subnet first.
5. Run the design checker and fix the issues.
6. Write the justification and export the report.

Key rules

- Usable hosts = $2^n - 2$ and must cover hosts + growth.
- A subnet starts on a multiple of its block size; no overlaps.
- Gateway = first usable address.
- Copper ≤ 100 m, fibre beyond. Phones, APs and cameras need PoE.
- Guests in their own VLAN, filtered at the firewall.

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